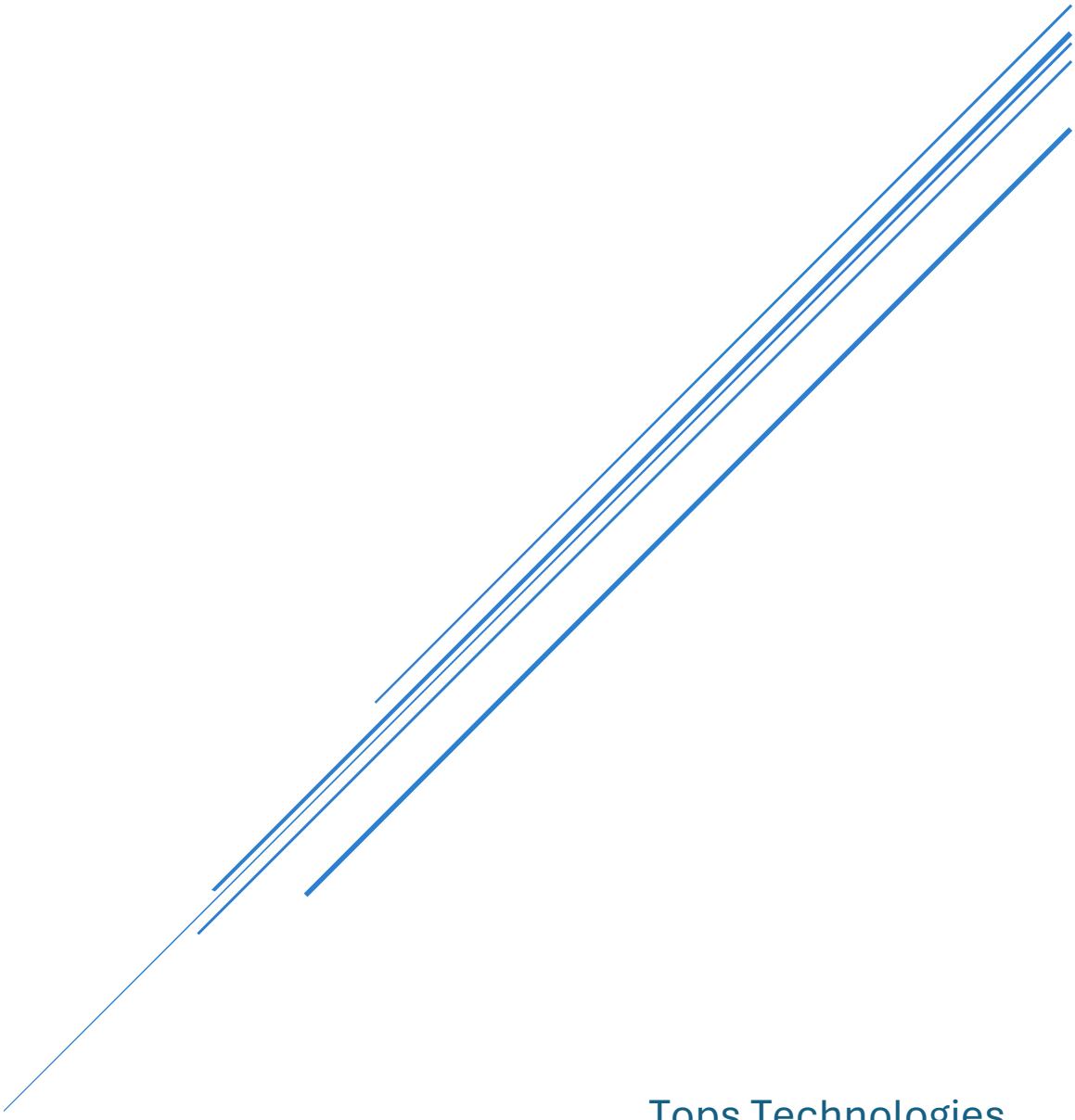
WIRELESS NETWORKING

Assignment - 11



Tops Technologies
Cyber Security

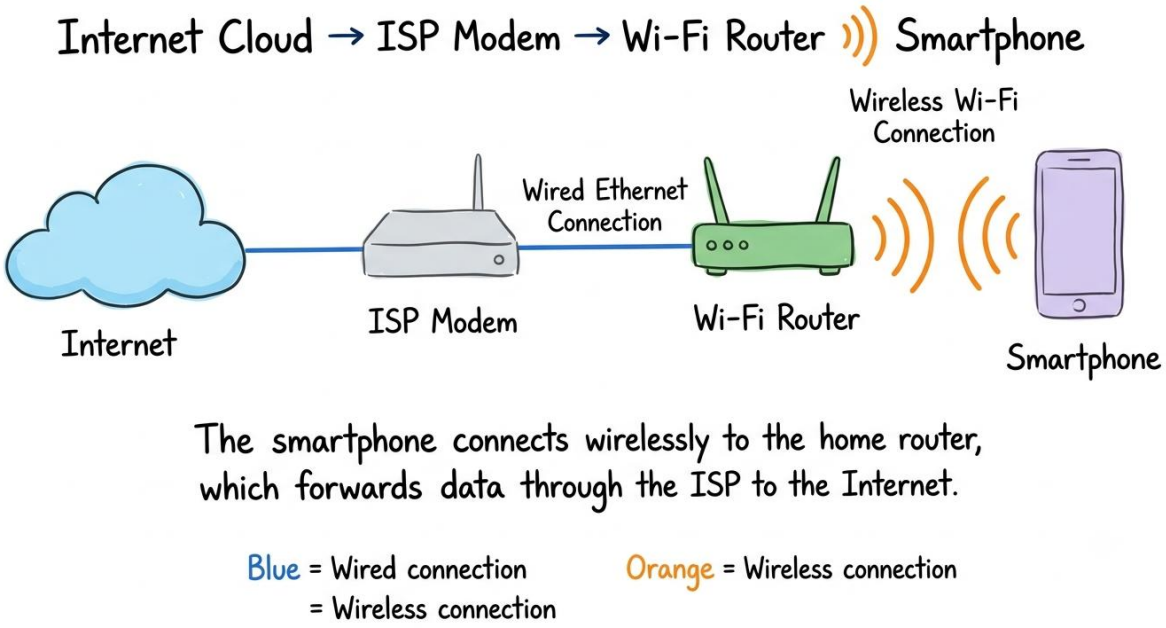
1. **List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.**

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Wi-Fi Standard	Frequency Band	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Speed up to 600 Mbps.
802.11ac (Wi-Fi 5)	5 GHz	Speed up to about 6.9 Gbps.
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz	Speed up to about 9.6 Gbps and better performance with many connected devices.

2. **Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.**

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1. **The smartphone connects wirelessly to the home Wi-Fi router.**
2. **The router sends and receives data through the ISP modem using a wired Ethernet connection.**
3. **The modem connects to the Internet Service Provider (ISP), providing access to the internet.**

3. **Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.**
   Hint: Consider both signal strength and security when choosing.

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Wi-Fi Network Scanning Activity

Sample observation table (example data — scan your own laptop or phone to record actual results).

No.	SSID (Wi-Fi Name)	Signal Strength	Security Type
1	Home_WiFi	Excellent (99%)	WPA2-Personal
2	Neighbour_WiFi	Medium (60%)	WPA2-Personal
3	Office_WiFi	Weak (35%)	WPA3-Personal

Which network would I choose and why?

I would choose Home_WiFi because it has excellent signal strength and uses WPA2 security. It should provide a stable and secure connection for browsing, online classes, and streaming, assuming it is my trusted home network with a strong password.

4. **Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.**

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SSID Hiding in Wi-Fi Networks

SSID hiding means disabling the broadcast of a Wi-Fi network's name so it does not normally appear in the list of available Wi-Fi networks.

- **Advantage: It makes the Wi-Fi network less visible to casual users who are searching for available networks.**
- **Disadvantage: It does not provide strong security because people can still discover a hidden network using network-scanning tools. Users must also enter the network name manually to connect.**